Preprocessing PASEC2019 Survey Data for Minimum Proficiency Risk Prediction: Strategy versus Implementation

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***Abstract*—This report audits the implementation of a preprocessing strategy for a binary classifier that flags Grade 6 students likely to be below the PASEC minimum proficiency level in mathematics, using only background questionnaire data. The PASEC2019 Grade 6 file (62,934 students, 749 columns, 14 countries) is cleaned of variable-specific missing-value codes, split 80/20 by school with country-by-label stratification, and transformed by a scikit-learn `Pipeline` whose parameters are learned on the training split only. Every planned step is compared with its execution in a decision log, and every decision is backed by a statistic and a diagnostic figure. Six changes were made during implementation, including capping before imputation, survey-weighted loss and metrics, and an elastic-net baseline; XGBoost and CatBoost run on the same pipeline. The pipeline reduces 749 raw columns to 32 model inputs with no student removed and no missing value left. Half of the label variance lies between schools, which shows why school-grouped splitting matters: ignoring schools inflates ROC AUC from 0.769 to 0.852. Borderline students and leave-one-country-out tests show where the background features stop being informative.**

***Index Terms*—data preprocessing, survey weights, missing data, data leakage, educational assessment, PASEC**

I. Introduction

SDG Indicator 4.1.1 measures the share of children who reach a minimum proficiency level in reading and mathematics. In 14 Francophone Sub-Saharan African countries this is measured by PASEC, coordinated by CONFEMEN [1]. A full assessment is expensive and periodic, but PASEC also records background information on every tested student. The project tests whether that information alone can flag a student as likely below the minimum in mathematics.

This report covers the preprocessing stage. Section II describes the data, label and split. Section III presents the pipeline. Section IV gives the decision log (Table I), Section V the statistical evidence behind each decision, and Section VI the analyses added after review of the strategy. Section VII explains the deviations and Section VIII verifies the output. The code, notebook and figures are at https://github.com/GentilleUwera/pasec-minimum-proficiency-risk. Every number below is printed by the notebook `notebooks/pipeline_execution.ipynb`.

II. Data, Label and Split

A. Raw matrix and missing-value codes

The raw Grade 6 file has $N = 62{,}934$ rows and $P = 749$ columns, from 3,298 schools. PASEC stores missing answers as codes whose meaning depends on the question (9 = Missing, 6 = Invalid, 7 = Not applicable, and longer forms such as 99, 999 or 9999 for numeric questions). They are converted to `NaN` variable by variable from the AFLEARN codebook [2], before the split, because the conversion uses no statistic of the data. In total 12,109,003 cells (25.69%) are missing; 12.35% of all cells are test items not administered by design.

B. Target label

PASEC reports mathematics as five plausible values per student. A student is labelled 1 when at least 3 of the 5 values are at or below 521.46, the boundary of the minimum proficiency level [1, Table 5.17]. This gives 39,685 positives (63.1%) and 23,249 negatives, a ratio of 1.71:1. The five per-value labels agree with Fleiss' $\kappa = 0.769$ [3], but 13,987 students (22.2%) are *borderline*: their values fall on both sides of the cutoff.

Two properties of the survey design matter for the rest of the work. First, plausible values are random draws from posterior distributions that PASEC conditions on the characteristics of students, of their teachers and classes, and of their principals and schools [1, Ch. 2, Sec. 3, p. 39], [4]. Part of the association between background features and the label is therefore inherited from the scaling model, not only from learning itself. This is a limitation of interpretation, not leakage, because no score or test item enters the features. Second, the final student weight `rwgt0` gives the number of students each sampled student represents [1, Ch. 1, Sec. 3.7, p. 24]. Weighted by `rwgt0`, the population share below the minimum is 69.1%, against 63.1% in the sample (Fig. 1.2).

C. Split

One fold of a five-fold `StratifiedGroupKFold` (seed 42) is held out as the test set. Groups are schools (country code plus `ID_ECOLE`, since the manual defines `ID_ECOLE` within a country), so classmates sharing teacher and principal answers never appear on both sides. Strata are the 28 country-by-label combinations. The result is 50,347 training students in 2,639 schools and 12,587 test students in 659 schools (20.0%),

with no shared school, label rates of 63.06% and 63.03%, and country shares within 0.04 percentage points.

III. Pipeline Architecture

Ten background features are selected: wealth index `ses`, age `qe62` and class size `qm22` (numeric); mother reads, father reads and grade repetition (binary); teacher training `qm5`, school location `qd31` and French at home `qe620` (ordinal); and country `ID_PAYS` (nominal). Test items, plausible values, identifiers and the 94 survey-design columns are excluded. Two fixed rules run before the split: a class enrolment of 0 becomes missing, and Yes/No is recoded to 1/0.

All learned steps sit inside one scikit-learn `Pipeline` [5] (`src/pipeline_builder.py`):

1) `Winsorizer` (logistic regression only): caps `ses` at mean ± 3 SD and `qe62`, `qm22` at $Q_1 - 1.5\,\mathrm{IQR}$ and $Q_3 + 1.5\,\mathrm{IQR}$;
2) `CountryImputer`: within-country mean (`ses`), median (`qe62`, `qm22`) or mode (binary, ordinal), plus a 0/1 flag per imputed feature;
3) `ColumnTransformer`: $\log(1 + x)$ and `StandardScaler` for numeric columns, `OrdinalEncoder` with explicit category order, and `OneHotEncoder` for country. The gradient boosting variants (XGBoost and CatBoost) skip capping, log and scaling; the CatBoost variant keeps country as one categorical column instead of 14 one-hot columns.

Fitted on `X_train` only, the pipeline stores capping bounds, 14 sets of country fill values, scaling means and standard deviations, and category lists, and applies them unchanged to `X_test`. Standardized training columns have mean 0 and SD 1; test columns have means within ± 0.04 and SDs between 0.992 and 1.022, as expected from parameters learned elsewhere. In cross-validation the whole pipeline is cloned and refitted inside each school-grouped fold (`src/evaluation.py`).

IV. Strategy versus Reality Decision Log

Table I compares each planned step with its execution. "Plan" refers to the preprocessing strategy report; "D" numbers refer to Section VII.

V. Statistical Evidence for Each Decision

A. Missingness mechanism and imputation

Only 33,317 of 50,347 training students (66.2%) have all ten features, so listwise deletion would discard a third of the sample. Little's test on the numeric block gives $\chi^2 = 329.5$ (df = 9, $p = 1.4 \times 10^{-65}$), rejecting MCAR [6]. Fig. 1.1 shows block missingness in the teacher questionnaire: class size is missing for 14.05% of training students and teacher training for 13.26%, together, and both depend on country (Cramér's V = 0.238 and 0.213, $p < 0.001$). Missingness of every feature is associated with country ($p < 10^{-40}$), which supports imputing within country rather than globally. The missing groups are also more often below the minimum (79.7% when `ses` is missing against 62.4% when observed; 82.6% against 61.7% for `qe620`), so MNAR cannot be ruled out. The 0/1 flags keep that signal for the model.

The fill statistic follows the observed skewness: mean for `ses` (skew -0.09), median for `qe62` (0.64) and `qm22` (3.32), mode for binary and ordinal answers. Two alternatives were run with the same folds and survey weights: iterative (MICE-style) imputation [7] and native missing-value handling in XGBoost. Neither changed performance (Table III), so the simpler within-country imputer is kept; the iterative version took about ten times longer.

B. Outliers and distribution shapes

Table II lists the distribution statistics that set each rule. The wealth index is near-symmetric (skew -0.09, excess kurtosis -0.56), so the Z-score rule applies. Age and class size are skewed, and class size is heavy-tailed (skew 3.32, kurtosis 26.82, maximum 352), so the IQR rule applies. With $Q_1 = 25$ and $Q_3 = 48$ (IQR = 23), the upper fence for class size is 82.5 students, which caps 1,680 values (3.88% of observed values). Fig. 2.1 shows the raw and capped distributions. Values are capped, not deleted: no row is removed, and capping is applied only in the logistic regression pipeline, where a class of 352 students would otherwise pull the coefficient. Gradient boosting receives the original values.

C. Collinearity and feature screening

The Spearman matrix (Fig. 3.1) adds two excluded candidates so the rule can be seen working. Two pairs pass $|\rho| \geq 0.85$: age and birth year ($\rho = -1.000$) and class size enrolled and present (`qm21`, $\rho = 0.887$, Pearson $r = 0.773$). With the candidates included, VIF reaches 1,946 for age and birth year, above the threshold of 10. Fig. 3.2 confirms the shape of both relations: birth year is an exact linear function of age (slope -1.00), and students present follows enrolment (slope 0.83). Applying the decision rule, `qm22` is kept over `qm21` (MI 0.0197 against 0.0138; 14.05% against 14.86% missing), and birth year is dropped (equal MI of 0.0075 and equal missingness, so the selected feature is kept). After removal the highest VIF is 1.87 for the features and 2.59 for the country dummies. Among the selected features the strongest pair is `ses` and `qd31` ($\rho = -0.54$).

Mutual information ranks the selected features (Fig. 3.3). It is used because it applies to numeric, ordinal and nominal variables alike. Country (0.080) and wealth (0.0735) carry most of the signal; each remaining feature is weak alone (at most 0.0197), so the model has to rely on combinations.

D. Transformations and scaling

Fig. 4.1 follows class size through four stages. $\log(1 + x)$ reduces skewness from 3.32 to -0.02 and excess kurtosis from 26.82 to 0.72. Min-Max scaling and standardization are linear, so they keep that shape and only change the range: Min-Max to [0, 1] and standardization to mean 0, SD 1. The pipeline uses log followed by standardization, because the elastic-net penalty treats all coefficients on one scale; Min-Max is shown

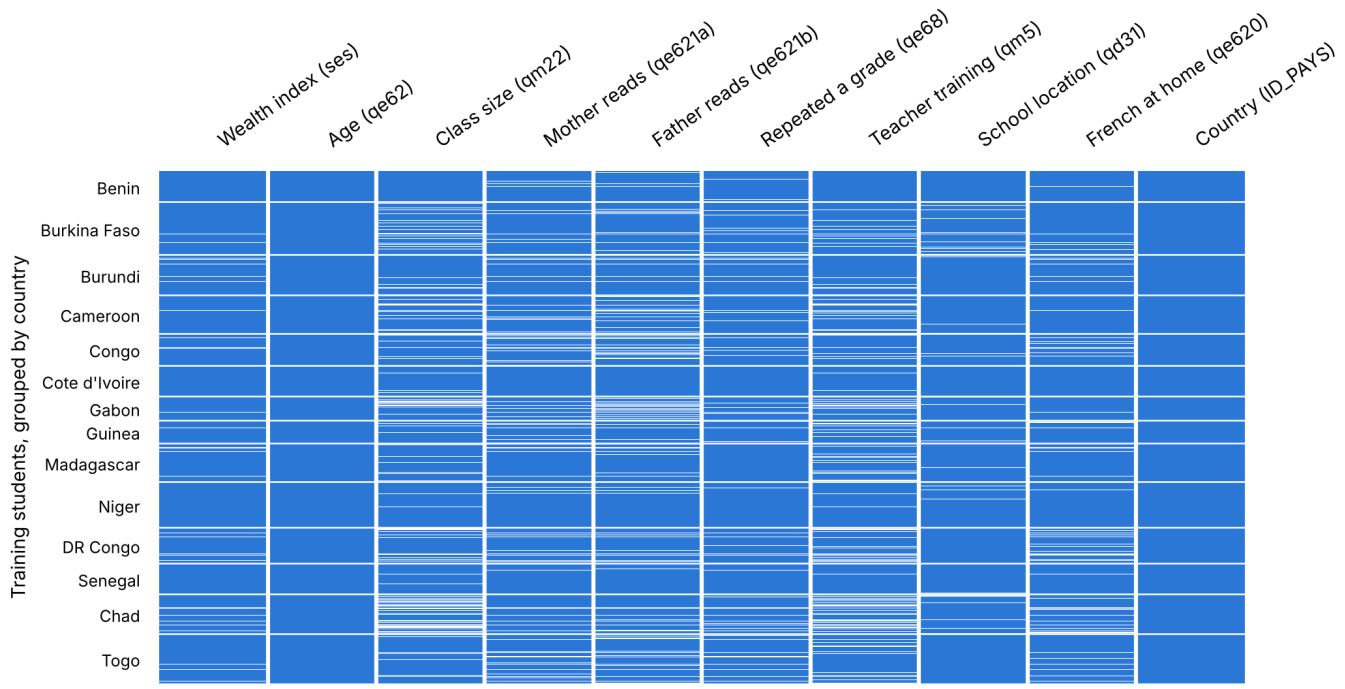


Fig. 1.1. Missingness pattern of the ten features, training set, students grouped by country (white = missing). Class size and teacher training are missing together in blocks within countries, the unlinked teacher questionnaire.

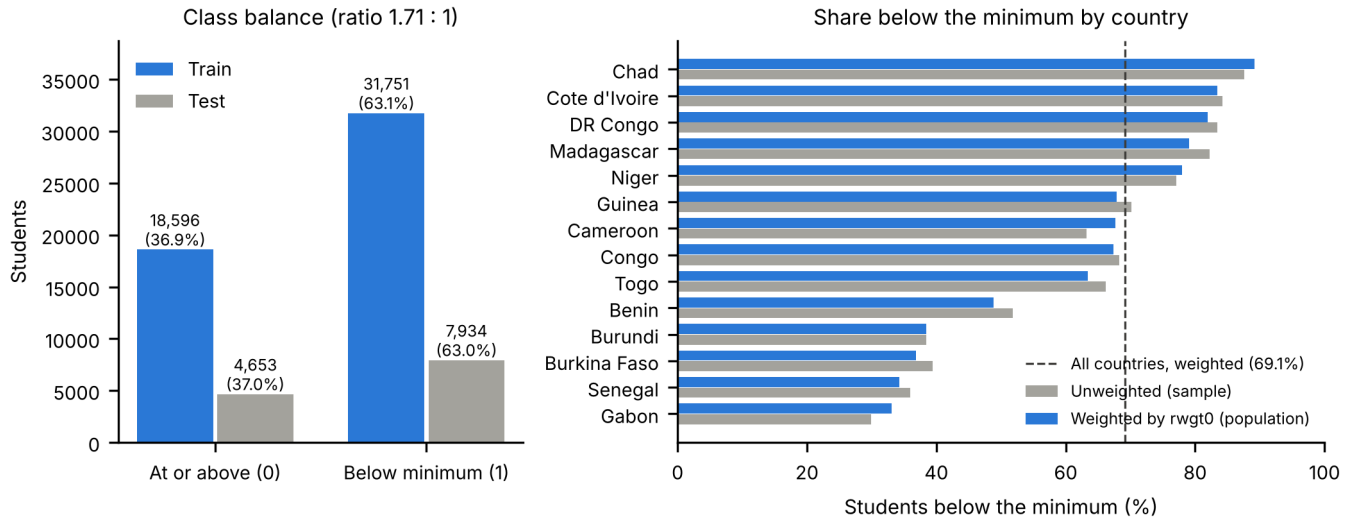


Fig. 1.2. Target distribution. Left: class balance in train and test (ratio 1.71:1 in both). Right: share below the minimum per country, unweighted and weighted by `rwgt0`. Weighting moves country shares by up to 4.5 points (Cameroon 63.2% to 67.7%) and the pooled share from 63.1% to 69.1%, because DR Congo carries the largest population weight.

for comparison only, since it is sensitive to the extreme values that remain in the test set.

E. Dimensionality reduction

PCA on the 32 standardized pipeline columns (Fig. 5.1) gives a first eigenvalue of 3.44 (10.76% of the variance), 14 components with eigenvalue above 1, and needs 23 components for 90% and 26 for 95% of the variance. The variance is spread widely, consistent with the low VIFs, so PCA would save few columns at the cost of interpretability. It remains a diagnostic; the model keeps the original features.

TABLE I
DECISION LOG: PLANNED STRATEGY AGAINST EXECUTED PIPELINE

Step	Planned strategy	Executed and measured	Evidence
Missing-value codes	Per-variable codebook map to NaN, before the split	12,109,003 cells (25.69%) converted; same as the audit	Notebook Sec. 1
Label	Majority of 5 plausible values ≤ 521.46	39,685 / 23,249; 13,987 borderline (22.2%)	$\kappa = 0.769$; Fig. 1.2
Split	80/20 `StratifiedGroupKFold`, groups = school, strata = country $\times$ label	50,347 / 12,587; 0 shared schools; label rate 63.06% / 63.03%	28 strata; ICC 0.497; Fig. 1.2
Missingness diagnosis	Little's test, missing vs observed label rates	MCAR rejected; country-linked block missingness in teacher items	$\chi^2 = 329.5$, df 9, $p = 1.4 \times 10^{-65}$; Cramér's V up to 0.238; Fig. 1.1
Imputation	Within-country mean / median / mode, global fallback	As planned; fallback never used (smallest country count 1,786)	Skew -0.09, 0.64, 3.32 set mean vs median; Fig. 1.1
Missing flags	One flag per imputed feature	9 flags; label rate 82.6% when `qe620` missing vs 61.7% observed	Notebook Sec. 5
Imputation check	MICE and native XGBoost NaN as sensitivity checks	Same weighted balanced accuracy (0.717 / 0.713)	Table III
Outlier rule	$\lvert Z\rvert > 3$ for `ses`; 1.5 IQR for `qe62`, `qm22`	Bounds [19.10, 78.64], [9, 17], [-9.5, 82.5]; 34, 540, 1,680 capped; 0 rows removed	Kurtosis 26.82 for `qm22`; Fig. 2.1
Capping order	Imputer, then winsorizer	Winsorizer moved before the imputer (D1)	Upper bound 82.5 vs 72.0
Feature ranking	Mutual information (MI) for every feature	Country 0.080, `ses` 0.0735, `qm22` 0.0197; small shifts from the plan (D2)	Fig. 3.3
Collinearity	Spearman $\lvert\rho\rvert \geq 0.85$, VIF > 10; drop lower MI, tie rules	`qe61y` and `qm21` dropped; tie made operational (D3)	$\rho = -1.000$, 0.887; VIF 1,946 $\rightarrow$ 1.87; Figs. 3.1, 3.2
Encoding	Ordinal integers; one-hot country	As planned; 14 country columns kept after a leave-one-country-out test	Table V
Transformation	$\log(1 + x)$ for class size, then StandardScaler	Skew 3.32 $\rightarrow -0.02$; kurtosis 26.82 $\rightarrow$ 0.72	Fig. 4.1
Class imbalance	`class_weight='balanced'`	Survey weight $\times$ class-balance factor in the loss (D5)	Table III
Survey weights	Report test metrics unweighted and weighted	Weighted and unweighted metrics in every check	Fig. 1.2; Table III
Models	L2 logistic regression and XGBoost	Elastic-net logistic regression (D4), XGBoost and CatBoost on the same folds	Table III
Nesting	Group by school	Random-intercept variance split; folds ignoring schools as a control	AUC 0.852 vs 0.769; Table VI
Borderline students	Noted as a limitation	Scored separately; probabilities by number of votes	Table IV; Fig. 6.1
PCA	Diagnostic only	23 of 32 components for 90%, 26 for 95%; not applied	Eigenvalues; Fig. 5.1
Leakage control	Full pipeline refit in each fold	School-grouped CV, pipeline cloned per fold	Weighted BA 0.705 to 0.729 across folds
Reproducibility	Pinned packages	scikit-learn 1.9.1 pinned, split sizes asserted (D6)	50,273 vs 50,347 rows with 1.7.2

TABLE II
OUTLIER STATISTICS ON THE TRAINING SET

	`ses`	`qe62`	`qm22`
Rule	$\lvert Z\rvert > 3$	1.5 IQR	1.5 IQR
Mean (SD)	48.87 (9.92)	12.85 (1.71)	39.31 (22.98)
Skewness	-0.09	0.64	3.32
Excess kurtosis	-0.56	1.51	26.82
Q_1 / Q_3	42.19 / 56.40	12 / 14	25 / 48
Bounds	[19.10, 78.64]	[9, 17]	[-9.5, 82.5]
Capped (% observed)	34 (0.07)	540 (1.08)	1,680 (3.88)

VI. ANALYSES ADDED AFTER STRATEGY REVIEW

A. Survey weights in the loss and the metrics

PASEC is a stratified cluster sample, so unweighted metrics describe the sample, not the population. Each training student's loss weight is `rwgt0` rescaled to mean 1, multiplied by a factor that gives both classes the same total weight. Metrics are reported both ways. Table III shows that weighting the loss moves the operating point (recall rises to 0.760 when measured at population level) while balanced accuracy stays near 0.72 and ROC AUC near 0.77. Per-fold weighted balanced accuracy for the elastic-net model ranges from 0.705 to 0.729.

B. Borderline students

Table IV scores the out-of-fold predictions separately. On the 11,210 borderline training students, weighted balanced accuracy falls to 0.557 and ROC AUC to 0.572, against 0.759 and 0.810 for students whose five values agree. The models are not less confident on borderline students (mean $\lvert p - 0.5\rvert$ of 0.202 against 0.210), which means confidence alone cannot identify them. Fig. 6.1 shows that the mean predicted probability rises steadily with the number of plausible values below the cutoff (0.316 at none to 0.588 at five for the elastic

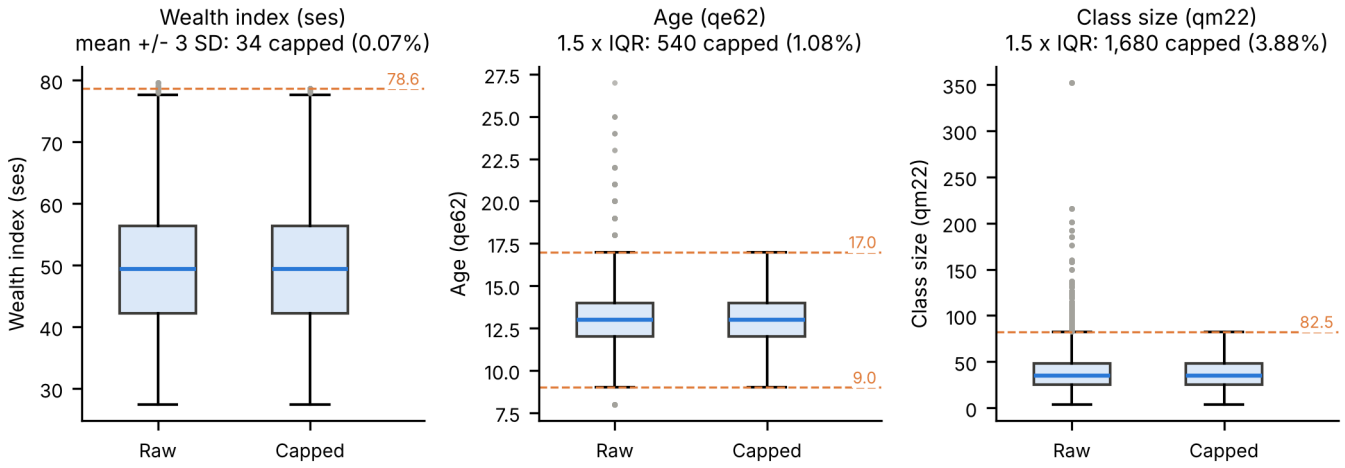


Fig. 2.1. Raw against capped values on the training set, with the learned bounds (dashed). Capping affects 0.07%, 1.08% and 3.88% of observed values; no student is removed.

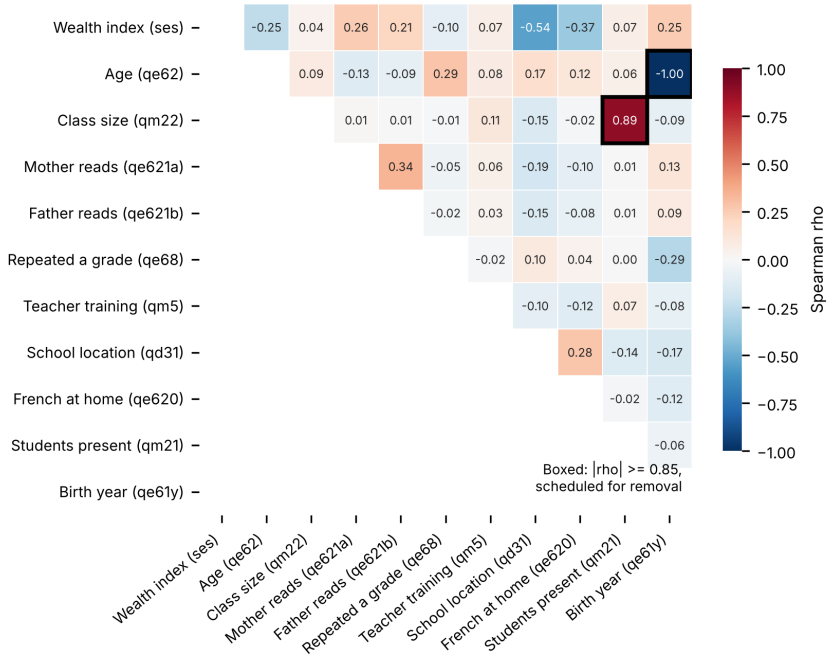


Fig. 3.1. Upper-triangle Spearman correlation matrix (imputed training data, with two candidate columns). Boxed cells pass $|\rho| \geq 0.85$ and are scheduled for removal.

net), so the ranking follows the underlying score even where the binary label is uncertain. Scoring the same predictions against each of the five per-value labels and averaging, as PASEC does for statistics on plausible values, gives a weighted balanced accuracy of 0.712 for the elastic net, 0.707 for XGBoost and 0.708 for CatBoost, against 0.717, 0.713 and 0.713 with the majority-vote label, so the label construction does not inflate the results.

C. *Country representation for new countries*

Country is the strongest single predictor (MI 0.080), but a one-hot column carries nothing for a country absent from training. A leave-one-country-out test trains on 13 countries and scores the 14th, for each country, with four representations (Table V). Two alternatives describe a country with continuous columns: country averages of the survey answers themselves (wealth, class size, repetition, location, home language; computed without labels), and the same plus government expenditure on education as a share of GDP from the World Bank [8] (2019; 2017 for DR Congo). Performance on an unseen

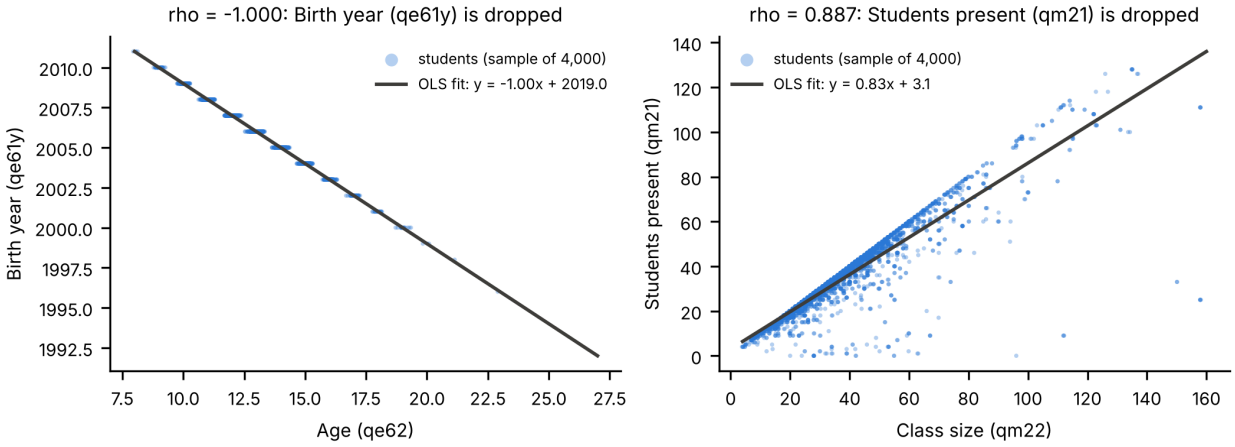


Fig. 3.2. The two collinear pairs with fitted least-squares lines (observed training values, samples of 4,000). The dropped column is named in each panel title.

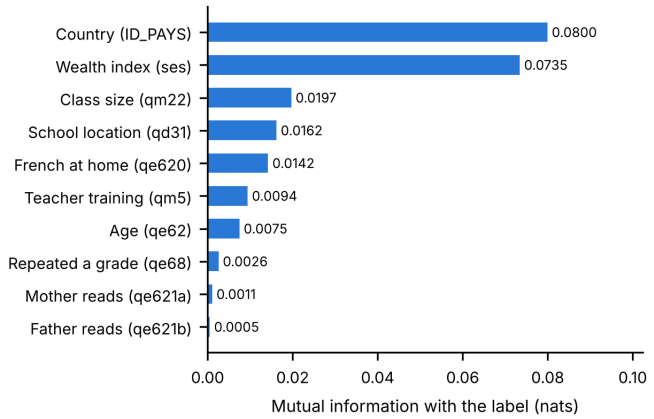


Fig. 3.3. Features ranked by mutual information with the label (training set).

TABLE III
SCHOOL-GROUPED 5-FOLD CV ON THE TRAINING SET

Set-up (metric weights)	Recall	F1	BA	AUC
LogReg, balanced (sample)	0.710	0.762	0.724	0.777
LogReg, balanced (`rwgt0`)	0.816	0.822	0.719	0.765
Elastic net, weighted (sample)	0.634	0.723	0.713	0.774
Elastic net, weighted (`rwgt0`)	0.760	0.798	0.717	0.763
XGBoost, weighted (sample)	0.661	0.738	0.718	0.782
XGBoost, weighted (`rwgt0`)	0.767	0.799	0.713	0.769
CatBoost, weighted (sample)	0.663	0.738	0.718	0.778
CatBoost, weighted (`rwgt0`)	0.772	0.802	0.713	0.762
Imputation sensitivity, weighted fit and metrics				
Elastic net, iterative imputation	0.759	0.798	0.717	0.764
XGBoost, native NaN	0.769	0.801	0.713	0.764

country drops from about 0.72 balanced accuracy within known countries to between 0.56 and 0.60 for every representation. The country-level columns match one-hot on ROC AUC (0.656 against 0.657) but lower balanced accuracy (0.563 and

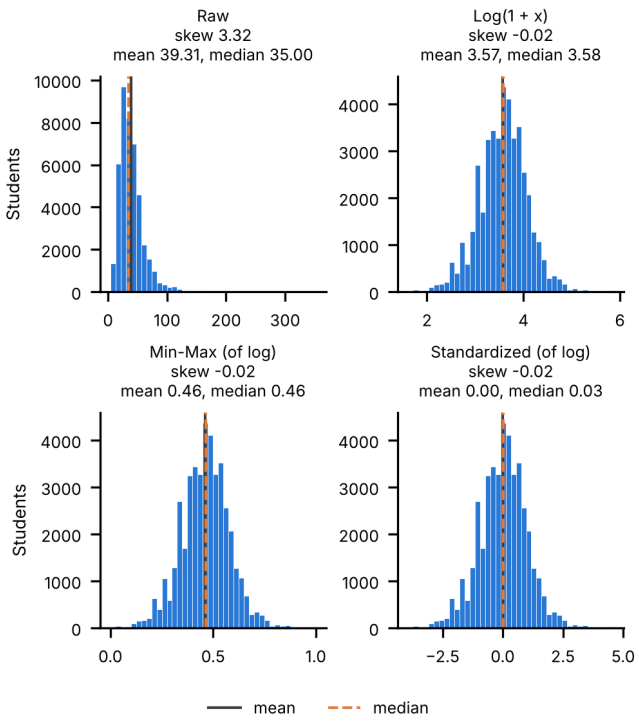


Fig. 4.1. Class size (`qm22`) through the transformation stages, observed training values.

0.570 against 0.601), because with only 13 training countries the model cannot separate a country effect from noise. One-hot encoding is therefore kept, and `CountryContext` stays in `src/` for a deployment that must score new regions.

D. Students nested in schools

A two-level random-intercept model (statsmodels `MixedLM`, linear probability, school intercepts) splits the label variance between and within schools (Table VI). With

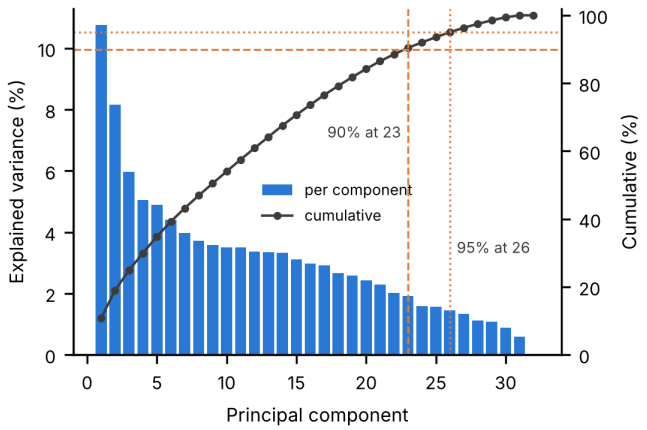


Fig. 5.1. PCA scree and cumulative variance of the 32 standardized training columns, with the 90% and 95% cutoffs.

TABLE IV
BORDERLINE AGAINST CONSISTENT STUDENTS (WEIGHTED)

Model	Students	n	BA	AUC	$\lvert p - 0.5 \rvert$
Elastic net	borderline	11,210	0.557	0.572	0.202
	consistent	39,137	0.759	0.810	0.210
XGBoost	borderline	11,210	0.556	0.568	0.228
	consistent	39,137	0.755	0.819	0.247
CatBoost	borderline	11,210	0.556	0.568	0.234
	consistent	39,137	0.755	0.810	0.255

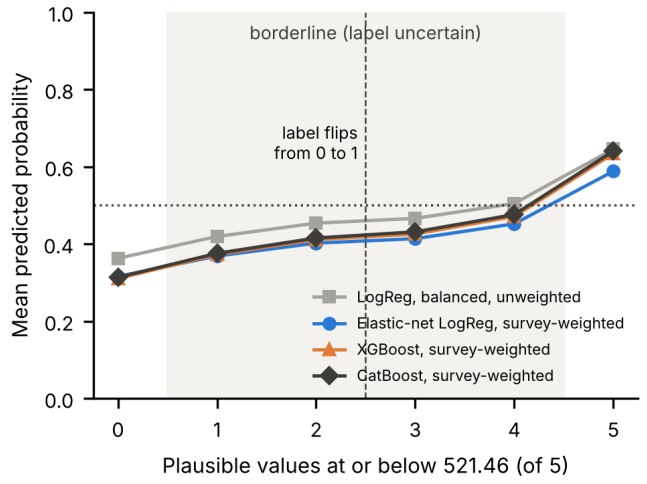


Fig. 6.1. Mean out-of-fold predicted probability by the number of plausible values at or below 521.46. The shaded band marks borderline students.

no predictors the intraclass correlation is 0.497: half of the variance in being below the minimum lies between schools. Country fixed effects bring it to 0.408, and all pipeline features only to 0.380; the features explain 12.6% of the between-school variance left after country. Two consequences follow. First, school context that the questionnaire does not measure limits how well any model can do on these

TABLE V
LEAVE-ONE-COUNTRY-OUT, MEAN OVER 14 COUNTRIES

Country representation	BA	AUC
None	0.583	0.636
One-hot	0.601	0.657
Survey context	0.563	0.656
Survey context + spending	0.570	0.656

TABLE VI
LABEL VARIANCE BETWEEN AND WITHIN SCHOOLS (TRAINING SET)

Fixed part	Between	Within	ICC
None	0.1152	0.1168	0.497
Country	0.0807	0.1168	0.408
Country + features	0.0705	0.1151	0.380

features, which is consistent with all four set-ups reaching about 0.72 balanced accuracy. Second, splitting classmates across folds would leak that context: repeating the XGBoost cross-validation with folds that ignore schools raises weighted ROC AUC from 0.769 to 0.852 and balanced accuracy from 0.713 to 0.776, an optimistic estimate that school-grouped folds avoid.

E. Model families

The three recommended families run on the same preprocessed data and folds (Table III). Elastic-net logistic regression [9] is the interpretable baseline. XGBoost [11] uses the 32-column matrix without scaling. CatBoost [10] receives 19 columns, with country as a native categorical feature instead of 14 one-hot columns. All three reach a weighted balanced accuracy of 0.713 to 0.717 and a weighted ROC AUC of 0.762 to 0.769, so at this stage the choice of model matters far less than the information in the features. The multilevel structure is described by the variance split above; a predictive multilevel GLM gives a new school a random intercept of zero, so for unseen schools it reduces to its fixed part, and it is kept for the modelling stage, where the processed metadata (school key, country, `rwgt0`) already supports it.

VII. DEVIATIONS AND RATIONALE

D1: capping before imputation. With the planned order, bounds were learned partly from filled-in values. Median filling of 14.05% missing class sizes narrowed the IQR, moving the upper fence from 82.5 to 72.0 and capping 2,845 values instead of 1,680 (the `ses` bound moved from 78.64 to 78.13). The `Winsorizer` now runs first, so bounds come from observed answers only.

D2: mutual information values. The plan computed MI after a global median fill; the pipeline computes it after within-country imputation. The order at the top is unchanged, with small shifts (`ses` 0.069 to 0.0735, `qm22` 0.015 to 0.0197). VIFs moved from 1.83 / 2.56 to 1.87 / 2.59 for the same reason.

TABLE VII
MATRIX DIMENSIONS AT EACH STAGE

Stage	N	P	Missing cells
Raw file	62,934	749	12,109,003
Selected features	62,934	10	44,283
Output, logistic regression	62,934	32	0
Output, XGBoost	62,934	32	0
Output, CatBoost	62,934	19	0
of which train / test	50,347 / 12,587	32	0

D3: operational tie rule. The plan's tie rule (more missing, then less interpretable) had no tolerance, and age and birth year differ in MI only by estimation noise. A tie is now an MI difference below 0.001, and the final tie-breaker keeps the selected feature instead of a subjective interpretability call.

D4: elastic-net baseline and a third model. The plan used an L2 penalty and XGBoost only; CatBoost was added after review. Following review, the baseline uses an elastic-net penalty (L1 ratio 0.5, $C = 1$), which can zero out weak features such as parental literacy (MI $\leq$ 0.0011) while handling mild correlation.

D5: survey-weighted loss. The plan used `class_weight='balanced'` and weighted only the reported metrics. The loss is now weighted by `rwgt0` times a class-balance factor, so the fit reflects the population (69.1% below the minimum) without being dominated by the majority class. The strategy's variable inventory also described `rwgt0` as a replicate weight; the manual defines it as the final student weight, with `rwgt1` to `rwgt90` as replicates.

D6: pinned library version. scikit-learn 1.6.1 and 1.7.2 assign schools to folds differently from 1.8 and 1.9, which gave a different split (50,273 / 12,661). Version 1.9.1 is pinned, and the notebook asserts the split sizes. The repository was also reorganized into `src/`, `notebooks/`, `figures/`, `data/` and `reports/`.

VIII. PIPELINE OUTPUT VERIFICATION

Table VII tracks the matrix through the pipeline. The 749 raw columns are reduced to 10 selected features and expanded to 32 model columns: 9 encoded features, 9 missing-value flags and 14 country columns. No information needed for the task is lost: all 62,934 students reach the output; the removed columns are test items and plausible values (target leakage), identifiers, survey-design variables, or questions outside the selected set; the two collinear columns are near-duplicates of kept features (Fig. 3.2); and the flags record every imputed value. The logistic regression and XGBoost variants produce 32 columns and the CatBoost variant 19 (country stays one categorical column), all with no missing cell on train and test.

IX. LIMITATIONS AND NEXT STEPS

The label inherits PASEC's conditioning on background variables, so feature importances describe the survey's scaling model as much as learning. Borderline students and new countries are the two weak points, and both will be reported in the modelling stage alongside test-set metrics, weighted and unweighted, with standard errors from the replicate weights. The test set has not been used in any decision in this report.

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